SCHOOL OF DATA SCIENCE AND FORECASTING

PROGRAM CODE: DS7A BATCH: 2024-26

PROGRAM TITLE: MASTER OF TECHNOLOGY (M.Tech.)-DATA SCIENCE

FIRST SEMESTER

Code	Title	Credits (L-T-P)
CORE COURSES		
DS7A-701	Statistical Research Methods	4 (3-1-0)
DS7A-703	Forecasting Methods	4 (3-1-0)
DS7A-705	Design and Analysis of Algorithms	4 (2-1-2)
DS7A-707	Advance Database Management Systems	4 (2-1-2)
DS7A-709	Python for Analytics	3 (2-0-2)
DS7A-711	NoSQL Laboratory	3 (1-0-4)
DS7A-713	Advance Excel	3 (1-0-4)
ELECTIVE COURSES-Any One		
DS7A-721	Cloud Computing	3 (2-0-2)
DS7A-723	Soft Computing	3 (2-0-2)

DETAILED SYLLABUS:

DS7A-701: Statistical Research Methods Credits: 4 (3-1-0)

Objective:

This course will familiarize students with the rudiments of statistical theory and ready them for effective academic and professional practice in the field of research process of industrial, system and social science.

COURSE DESCRIPTION:

Unit I: Introduction to Research Methods and Measures of central tendency:

Meaning and Objectives of Research, Significance of Research. Data collection primary and secondary data. Different Sampling techniques, questionaries' and survey. Google Document.

Unit II: **Basis Statistics and Regression:**

Mean deviation, Standard deviation, Variance, Co-efficient of variation. Types of correlation, Methods of Correlation, Co-efficient of correlation, Properties of correlation, Rank Correlation. Difference between correlation and regression, Regression Lines, Regression Equations.

Unit III: Testing of Hypothesis:

Procedure of Testing Hypothesis, Standard Error and Sampling distribution, Estimation, Student's t-distribution, Chi-Square test and goodness of fit, F-test and analysis of variance. Factor analysis.

Unit IV: Statistical Quality Control:
Introduction and Process Control, Control Charts for X and R, Control Charts for X and S., p chart np chart, c chart. Software for statistical analysis - SPSS, R, MS Excel.

Text Books:
S.P. Gupta: Statistical Method, S. Chand
C.K. Kothari: Research Methodology, New Age International

Course Outcomes:
- To be able to understand the basic concept of statistics and data collection.
- To be able apply advanced knowledge in statistics to experimental and applied research
- To be able to understand the concepts of validity and probability as they apply to different set of data.
- To be able to critically evaluate the methodological designs and select appropriate analytical strategies for their research projects.
- To understand the interpretation and appropriate reporting requirements for statistical and data analysis.
- To be able to use statistical packages required quantitative analysis (e.g., R, SPSS and Excel).

DS7A-703: Forecasting Methods Credits: 4 (3-1-0)

Objective: This subject is designed in such a way to provide the basic concepts of quantitative and qualitative forecasting methods, risk and uncertainty in forecasting, cause and effect relation on different dynamic situations.

Course Description:
Unit I: Introduction:
Forecasting perspective, alternatives to forecasting, types of forecasting, an overview of types of forecasting methods, basic steps in forecasting. Common time series pattern, graphical and numerical summaries. Measuring forecasting accuracy: Mean, Median, Mode, Error Standard Deviation, MAD, ME, MSE, MAE prediction intervals, transformations and adjustments. Principles of decomposition, moving averages, local regression smoothing, classical decomposition, Census Bureau methods, STL decomposition, forecasting and decomposition. Correlation measurement, Auto-correlation, ACF, Durbin-Watson statistics.

Unit II: Methods and Models:
Forecasting scenarios, averaging methods: mean, moving averages; exponential smoothing methods-single exponential smoothing, ARRSES, Holt’s linear method, Holt-Winter’s trend and seasonality method, comparison of methods, general aspects of smoothing methods.

Simple regression, forecasting with simple regression, non-linear relationships. Regression with time series, regression and forecasting, econometric models for forecasting. Univariate ARIMA model: examining stationary, ARIMA models, forecasting with ARIMA models, ARIMA application.

Unit III: Qualitative forecasting: Its meaning and need, stages of innovation. Delphi: advantages and disadvantages of committees, Delphi procedure, conducting a Delphi based study. Forecasting by analogy. Growth curves-substitution curves, Pearl curve, Gompertz curve, Fisher-Pry curve, selection of proper growth curve, estimation of upper limit.

Unit IV: Measures of technology-scoring models, Causal models, technology only models, techno-economic models. Normative methods-relevance trees, morphological models, mission flow diagrams. Uses of forecasting methods in practice, advantages and limitation of forecasting. Case Studies.

Text Books:

1. Spyros Makridakis: Forecasting Methods and application Wiley
2. N.P. Nagpal: Forecasting Techniques, RBSA
3. Stephen A. Delurgio: Forecasting Principles and Application,McGraw Hill
4. Spyros Makridakis: Forecasting Methods and Application, Wiley publication
5. Joseph P. Martino: Technology forecasting for decision making, McGraw-Hill engineering and technology management series
6. Stephen A. Delurgio: Forecasting principles and applications, Irwin McGraw-Hill

Course Outcomes:

- Discuss the key factors which affect the success of forecasting procedures.
- Use Basic Statistical Techniques and statistical Graphics to forecast values.
- Find different sets of Smoothed or Average values to be used when forecasting.
- Understand the key concepts needed to use the Linear Regression model when forecasting.
- Model and Forecast the Seasonal component of a set of values.
- Model the different types of cyclical behaviour observed in different sets of values.
- Understand and use the Box-Jenkins or ARMA Procedure.
- Understand the ways of forecasting in the new emerging field of technology.
- To identify different causing variables affecting forecast and model them.
- Combine the trend, seasonal and cyclical components to produce a more accurate forecast of a set of values.

DS7A-705: Design and Analysis of Algorithms Credits: 4 (2-1-2)

Objective: This course is designed to equip students with a comprehensive understanding of both fundamental and advanced algorithmic concepts. The course discusses the core principles of algorithms, exploring mathematical foundations, algorithm analysis, and various algorithmic strategies such as divide-and-conquer, dynamic programming, and greedy algorithms. By the end of the course, students will be proficient in selecting and implementing appropriate algorithmic approaches, making informed decisions to optimize performance and solve problems with confidence.

COURSE DESCRIPTION:

UNIT I: The efficient algorithm, Average, Best and worst case analysis, Amortized analysis, Asymptotic Notations, Analyzing control statement, Loop invariant and the correctness of the algorithm, Algorithms and Flowcharts, Static and Dynamic Memory Allocation, Arrays, Pointers and Strings, Multidimensional Arrays, Pointer to Structure, Library Functions of Strings, Sorting Algorithms and analysis: Bubble sort, Selection sort, Insertion sort, Shell sort Heap sort, Sorting in linear time: Bucket sort, Radix sort and Counting sort.

UNIT II: Stacks and Queue, Circular Queue, De-queue and Priority Queue, Singly Linked List, Doubly Linked List, Circular Linked List, Binary Search Trees, Tree Traversal, Threaded Binary Tree, AVL Tree B Tree, B+ Tree, Recurrence and different methods to solve recurrence, multiplying large Integers Problem, Problem Solving using divide and conquer algorithm - Binary Search, Max-Min problem, Sorting (Merge Sort, Quick Sort), Matrix Multiplication, Exponential.

UNIT III: General Characteristics of greedy algorithms, Problem solving using Greedy Algorithm - Activity selection problem, Elements of Greedy Strategy, Minimum Spanning trees (Kruskal's algorithm, Prim's algorithm), Graphs: Undirected Graph, Directed Graph, Traversing Graphs, Shortest paths, The Knapsack Problem, Job Scheduling Problem, Huffman code.

UNIT IV: Backtracking and Branch and Bound: The Eight queens' problem, Knapsack problem, Travelling Salesman problem, Minimax principle. String Matching: Introduction, The naive string matching algorithm, The Rabin-Karp algorithm, String Matching with finite automata. NP-Completeness: The class P and NP, Polynomial reduction, NP-Completeness Problem, NP-Hard Problems. Travelling Salesman problem, Hamiltonian problem, Approximation algorithms.

Text Books:

1. Levitin, A., Introduction to the design and analysis of algorithms. Pearson Education Asia.
2. Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C., Introduction to algorithms (2nd ed.). PHI Learning Pvt. Ltd./Pearson Education.
3. Lee, R. C. T., Tseng, S. S., Chang, R. C., & Tsai, T., Introduction to the design and analysis of algorithms: A strategic approach. McGraw-Hill.
4. Horowitz, E., Sahni, S., & Rajasekharam, S., Fundamentals of computer algorithms. Galgotia Publications Pvt. Ltd.
5. Dave, P. H., & Bhalchandra, H., Design and analysis of algorithms. Pearson Education.

DS7A-707: ADVANCED DATABASE MANAGEMENT SYSTEMS Credits: 4 (2-1-2)

Objective: The Advanced Database Management Systems course is intended to explore both fundamental principles and advanced features of the classical and modern Database Management Systems. Students will engage with topics such as relational database design and implementation, distributed databases, and emerging trends in database technology. The course is structured into four modules, providing a comprehensive understanding of DBMS. By the end of the course, students will have gained the knowledge necessary to design and implement sophisticated database systems, enhancing their ability to manage and utilize information effectively.

COURSE DESCRIPTION:

UNIT I: Overview of DBMS: Comparison between Database approach and Traditional file accessing approach, Advantages of database systems, Schemas and instances, Data Dependency, Data Dictionary, and Meta Data. Data models, Types of Data models (Object Oriented, Record Based and Physical data models), E-R Modelling.

UNIT II: Relational Data model: Domains, Tuples, Attributes, Keys, Relational database, Schemas, Integrity constraints, Relational algebra and relational calculus; Normalization: Normal forms (INF, 2NF, 3NF, BCNF), Functional dependency, Decomposition, Dependency preservation and lossless join.

UNIT III: Structured Query Language: DDL, DML, DCL, TCL, SQL Functions, integrity constraints, various joins, sub-query, index, View, Sequence, and Clusters. Transaction Management, ACID properties, Concurrency Control, locking based algorithms, timestamp ordering algorithms, deadlock management.

UNIT IV: Object Oriented, Temporal and Spatial Databases: Object Oriented Databases, ODMG Model, Object Definition Languages (ODL), Object Query Languages (OQL), Temporal Databases, Time ontology, structure, and granularity, Temporal data models, Spatial Databases, Types of spatial data, Geographical Information Systems (GIS), Conceptual Data Models for spatial databases, Logical data models for spatial databases, Physical data models, Clustering, Space filling curves, Storage methods, R-tree, Query processing.

Text Books:

1. Silberschatz, A., Korth, H., & Sudarshan, S., Database System Concepts (5th ed.). McGraw-Hill.
2. Elmasri, R., & Navathe, S. B.m Fundamentals of Database Systems. Benjamin Cummings Publishing Company.
3. Rob, P., & Coronel, C., Database Systems (7th ed.). Cengage Learning.
4. McFadden, F. R., Hoffer, J. A., & Prescott, M. B., Modern Database Management (5th ed.). Pearson Education Asia.

5. Bayross, I., SQL, PL/SQL: The Programming Language of Oracle (4th revised ed.). BPB Publications.
6. Yeung, A. K. W., & Hall, G. B., Spatial Database Systems: Design, Implementation and Project Management. Springer.

DS7A-709: Python for Analytics Credits: 3 (2-0-2)

Objective: The main objective is to help students to understand the fundamentals of python. Student will learn how to analysis data using Python.

COURSE DESCRIPTION:

UNIT I: Introduction to Python: Python versus Java, Python Interpreter and it's Environment, Python installation, Python basics: variables, operators, Strings, Conditional and Control Statements, loops; Data structures: lists and dictionaries; functions: global functions, local functions, lambda functions and methods.

UNIT II: Object Oriented Programming Concepts: Class, object, constructor, destructor and inheritance; Modules & Packages, File Input and Output, Catching exceptions to deal with bad data, Multithreading, Database Connectivity.

UNIT III: Numpy: Creating Arrays, Arrays Operations, Multidimensional ArraysArrays transformation, Array Concatenation, Array Math Operations, Multidimensional Array and its Operations, Vector and Matrix.

Visualization: Visualization with matplotlib, Figures and subplots, Labeling and arranging figures, Outputting graphics.

UNIT IV: Pandas: Manipulating data from CSV, Excel, HDF5, and SQL databases, Data analysis and modelling with Pandas, Time-series analysis with Pandas, Using Pandas, the Python data analysis library, Series and Data Frames, Grouping, aggregating and applying, Merging and joining.

Text Books:

1. McKinney Wes, "Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython", O'Reilly Media, 2012.
2. Hauck Trent, "Instant Data Intensive Apps with Pandas How-To", Packt Publishing Ltd, 2013.
3. Beazley David M.,"Advanced Python Programming", Pearson Education,2009.
4. Chun Wesley , Core Python Programming, 3rd Edition,Prentice Hall Professional, 2012.
5. Telles Matt "Python Power!: The Comprehensive Guide", Cengage Learning, 2008.
6. McKinney Wes & PyData Development Team, "pandas: powerful Python data analysis toolkit", Release 0.13.1, Feb 2014.
7. https://docs.python.org/3.4/tutorial/
8. http://www.tutorialspoint.com/python/python_quick_guide.htm

Course Outcomes:

- The student will learn core data types of python.
- The student will learn conditional and looping operations in python.
- The student will be able to work with Object-oriented concepts and Database connectivity in python.
- The student will be able to analyze data using Pandas and Numpy.
- The student will be able to visualize the data using seaborn and matplotlib.

DS7A-711 NoSQL Laboratory Credits: 3 (1-0-4)

Objective: This laboratory course will discuss the NoSQL databases and the characteristics that distinguish them from traditional relational database management systems. Core concepts of NoSQL databases will be presented, followed by an exploration of how different database technologies implement these core concepts. Students will engage in practical exercises that involve 1-2 databases from each of the four main NoSQL data models (key-value, column family, document, and graph), highlighting the business needs that drive the development and use of each database. By the end of the course, students will have gained hands-on experience with various NoSQL databases and will be able to make informed decisions about database technologies for specific business applications.

COURSE DESCRIPTION:

UNIT I: Introduction to NoSQL: Basics, Storage Architecture, Operations, Query Model, Modifying Data Stores and Managing Evolution, Indexing and Ordering Data Sets, Managing Transactions and Data Integrity.

UNIT II: Introduction to Key-Value Databases (Redis Lab), Document Stores (MongoDB Lab), Column Family Stores (Cassandra Lab), Graph Databases (Neo4j Lab). Choosing a NoSQL Database.

UNIT III: MongoDB functions, operators and types of operations in MongoDB, use of Indexing, Advanced Indexing, aggregation and Map Reduction.

UNIT IV: Make use of MongoDB commands and queries, aggregate pipelines to extract data, optimization of queries by creating indexes, Development of aggregate pipelines for text search in collections

Text Books:

1. Harrison, G., Next-Generation Databases. Apress. ISBN: 9781484213292.
2. Sullivan, D., NoSQL for Mere Mortals. Addison-Wesley Professional. ISBN: 0134023218.
3. Chodorow, K., MongoDB: The Definitive Guide (2nd ed.). O'Reilly Media.
4. Banker, K., et al., MongoDB in Action (2nd ed.). Manning Publications.

5. Sharma, M., MongoDB Complete Guide (1st ed.). BPB Publications.
6. Installation of MongoDB: https://www.youtube.com/watch?v=dEm2AS5amyA
7. Aggregation: https://www.youtube.com/watch?v=vx1C8EyTa7Y
8. MongoDB in Action: https://www.manning.com/downloads/529
9. MongoDB Exercises: https://www.w3resource.com/mongodb-exercises/

Suggested List of Programs to be Discussed in Lab:

1. Retrieve all movies from the 'movies' collection that have complete information and have received more than 1000 votes on IMDb.

2. Retrieve all movies from the 'movies' collection that have complete information and have an IMDb rating higher than 7.

3. Retrieve all movies from the 'movies' collection that have complete information and have a viewer rating higher than 4 on Tomatoes.

4. Retrieve all movies from the 'movies' collection that have received an award.

5. Find all movies with title, languages, released, runtime, directors, writers, countries from the 'movies' collection in MongoDB that have a runtime between 60 and 90 minutes.

6. Find all movies with title, languages, released, runtime, directors, writers, countries, imdb from the 'movies' collection in MongoDB for the top 5 movies with the highest IMDb ratings.

7. Find all movies from the 'movies' collection in MongoDB with the average runtime of movies released in each country.

8. Find from the 'movies' collection in MongoDB with the most common genre among the movies.

Click me to see the solution

9. Find the movies released in the year with the highest average IMDb rating from the 'movies' collection in MongoDB.

10. Find the top 10 directors with the most movies from the 'movies' collection in MongoDB.

11. Write a query in MongoDB to find the average IMDb rating for movies with different ratings (e.g., 'PG', 'R', 'G') from the 'movies' collection.

12. Write a query in MongoDB to find the oldest movie with an award win from the 'movies' collection.

13. Write a query in MongoDB to find the movie with the highest IMDb rating and viewer rating on Tomatoes from the 'movies' collection.

14. Write a MongoDB query to find the restaurants that do not prepare any cuisine of 'American' and their grade score more than 70 and latitude less than -65.754168.

15. Write a MongoDB query to find the restaurants which do not prepare any cuisine of 'American' and achieved a score more than 70 and located in the longitude less than -65.754168. Note : Do this query without using $and operator.

16. Write a MongoDB query to find the restaurants which do not prepare any cuisine of 'American' and achieved a grade point 'A' not belongs to the borough Brooklyn. The document must be displayed according to the cuisine in descending order.

17. Write a MongoDB query to find the restaurant Id, name, borough and cuisine for those restaurants which contain 'Wil' as first three letters for its name.

18. Write a MongoDB query to find the restaurant Id, name, borough and cuisine for those restaurants which contain 'ces' as last three letters for its name.

19. Find all listings with listing_url, name, amenities, host in the listingsAndReviewscollection that have a host with a Jumio verification and a about section.

20. Retrieve all documents with listing_url, name, host, price in the listingsAndReviews collection where the host_total_listings_count field is greater than 1.

21. Retrieve all documents with listing_url, name, property_type, bed, price in the listingsAndReviewscollectionwhere the property_type field is equal to "Apartment" and the beds field is greater than or equal to 2.

22. Find all listings with listing_url, name, property_type, bed, bathrooms, price in the listingsAndReviews collection that have a minimum of 2 bathrooms.

23. Find all listings with listing_url, name, property_type, bed, price, guests_included in the listingsAndReviews collection that have a maximum of 5 guests included in the price.

DS7A-713: Advanced Excel Credits: 3 (1-0-4)

Objective: The main objective of this course is to learn analysis of data using MS Excel, resulting in less time and better understands what the data means.

COURSE DESCRIPTION:

Unit I: Introduction to Excel User Interface**,** Application, Workbook, Worksheets & its Components, Named Ranges; Formatting: Cell Color, Font Color, Indents, Alignments, Number Formats, Custom Formats, Editing commands; Data Sorting: Built-in Sort, Sorting Levels, Custom Sort; Data Filtering: Auto Filter – Filter By Color, Filter by Icono Advanced Filter, Remove Duplicates; Data Subtotal – Built-In Subtotal (Nested Subtotal).

Unit II: Data Validation: Based on cell values (text length, whole no Based on Formulas, List Dropdown, Circle Invalid Data, Input & Error Messages; Data Grouping: Grouping Rows, Grouping Columns. Data Tables: Conditional Formatting, Formatting based on Cell values, Formatting based on Formulas, Icon Sets (bars, scales, icons), Freezing Panes, Text-to-Columns, Delimited, Fixed Length; Data Consolidation (from multiple files), Getting External Data into Excel, From MS Access, From Text files, From Web, Other Data Sources.

Unit III: Formulas, TEXT Functions, IF, ERROR Functions, LOGICAL Functions, VLOOKUP, HLOOKUP, COUNTIF, SUMIF, SUMPRODUCT, DATE & TIME FUNCTIONS, FORMULA TEXT, Information Functions (ISNA, ISEVEN, ISERR…).

Unit IV: Charts: Chart Types, Chart Components, Primary Vs Secondary Axis, Chart Formatting, Sparkline (2010 and above); Pivot Tables: Introduction & Creation, Slicer, TimeLine, Pivot Charts, Calculated Fields, Calculated Items, Grouping, Formatting – Number/Conditional, PowerPivot, PowerView.

Text Books:

1. John Walkenbac, “Excel 2016 Bible”, John Willey & sons.
2. Jordan Goldmeier , “Advanced Excel Essentials”, Apress Publisher.
3. Conrad George Carlberg , “Business Analysis with Microsoft Excel”, Que Publishers.
4. Bernd Held, ”Microsoft Excel Functions & Formulas”, Wordware publishing, Inc.
5. Steven Roman ,”Writing Excel Macros with VBA” O’Reilly Media.

Course Outcomes:

- The student will be able to perform basic operations in Excel.
- The student will be able to summarize data using Grouping and pivot table.
- The student will be able to write conditional statements and perform LOOKUP operations.
- The student will be able to create charts in excel.
- The student will be able to create a dashboard in excel.

DS7A-721: Cloud Computing Credits: 3 (2-0-2)

Objective: The main objective of this subject is to help student to understand Cloud and It’s Services, Architecture, Deployment, Core Issues, Strengths and limitations of cloud computing.

UNIT I: Introduction to Cloud Computing: Overview of Cloud Computing, History of Cloud Computing, Importance of Cloud Computing, advantages and disadvantages of Cloud Computing, Applications, Cloud computing vs. Cluster computing vs. Grid computing, Future of Cloud Computing; Cloud Computing Architecture: Cloud computing stack, Comparison with traditional computing architecture (client/server), Cloud Service Models(XaaS), Deployment Models (Public cloud, Private cloud, Hybrid cloud).

Unit II: Cloud Service Models and Virtualization: Infrastructure as a Service (IaaS): Introduction, Introduction to virtualization, Different approaches to virtualization, Hypervisors, Virtual Machine(VM), Resource Virtualization(Server, Storage, Network),VMware vSphere, Machine Image ,Porting Applications Case study on Amazon EC2;Platform as a Service(PaaS): Introduction to PaaS, advantages and disadvantages of PaaS, case study on Microsoft Azure; Software as a Service(SaaS): Introduction to SaaS, Web services, Web 2.0, Web OS; Development Services and Tools : Amazon Ec2, Google App Engine, IBM Clouds.

UNIT III: Capacity Planning: Defining Baseline and Metrics, Baseline measurements, System metrics, Load testing, Resource ceilings, Server and instance types, Network Capacity, Scaling. Understanding Service Oriented Architecture, Moving Applications to the Cloud, Working with Cloud-Based Storage, Working with Productivity Software.

UNIT IV: Managing the Cloud: Administrating the Clouds, Cloud Management Products; Cloud Security: Security Overview, Cloud Security Challenges and Risks, Security Governance, Risk Management: Security Monitoring, Security Architecture Design, Data Security, Application Security, Virtual Machine Security, Identity Management and Access Control, Authentication in cloud computing, Client access in cloud, Cloud contracting Model. Using the Mobile Cloud: Working with Mobile Devices and Working with Mobile Web Services;

Case Studies on Various Clouds: Google Web Services, Amazon Web Services, Microsoft Cloud Services, IBM Clouds, Eucalyptus.

Text Books:

1. Kris A Jamsa: Cloud computing : SaaS, PaaS, IaaS, Virtualization, Business Models, Mobile, Security and More 2013Jones & Bartlett Learning ISBN-13: 9781449647391.
2. Michael Miller, Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online, Que Publishing, August 2008.
3. Toby Velte, Anthony Velte, Robert Elsenpeter, “Cloud Computing, A Practical Approach”, TMH, 2009.
4. Sosinsky B., “Cloud Computing Bible”, Wiley India
5. Cloud Computing: Principles and Paradigms, Editors: Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, Wile, 2011.
6. Cloud Security: A Comprehensive Guide to Secure Cloud Computing, Ronald L. Krutz, Russell Dean Vines, Wiley-India, 2010.

Course Outcomes:

- The student will be able to understand the basics of Cloud Computing.
- The student will be able to understand Cloud Computing Models.
- The student will be able to work with AWS Cloud Platform.

- The student will be able to manage cloud platform.
- The student will be able to understand various cloud services.

DS7A-723: Soft Computing Credits: 3 (2-0-2)

Objectives:

1. An understanding of the concept of different soft computing tools.
2. All soft computing techniques will be more effective to solve the problem efficiently.
3. Develop the skills to gain a basic understanding of neural network theory and fuzzy logic theory.

Unit-I: Introduction to Neural Network**:** Concept of Neural Network , biological neural network, comparison of ANN with biological NN, evolution of artificial neural network, Basic models, Types of learning, Linear separability, McCulloch-Pitts neuron model, Hebb rule.

Unit-II: Supervised and unsupervised learning: Perceptron, Backpropagation algorithm, Adalines and Madalines. Counter-propagation networks, Adaptive Resonance Theory, Kohonen's Self Organizing Maps, Neocognitron, Associative memory, Bidirectional Associative Memory.

Unit-III: Fuzzy logic and fuzzy sets: fuzzy relations, fuzzy graphs, fuzzy arithmetic and fuzzy if-then rules, Process control using fuzzy logic, Decision-making fuzzy systems, Applications of fuzzy logic, Hybrid systems like neuro-fuzzy systems.

Unit-IV: Evolutionary Computation: Population-based Search, genetic algorithms and evolutionary computation, Swarm optimization, Ant colony optimization. Search techniques like Simulated Annealing, Tabu search etc.

TEXT BOOKS:

1. Soft Computing and Intelligent Systems Design by F.O. Karray and C.DeSilva,Pearson Publication.
2. Neural Networks, Fuzzy Logic and Genetic Algorithms by Rajsekaran and Pai, PHI Publication.
3. S.N. Shivnandam, "Principle of soft computing", Wiley.

Course Outcomes:

After the completion of this course students will be able to

1. Identify different neural network architectures, algorithms, applications and their limitations.
2. Analyze learning rules for each of the architectures and learn several neural network paradigms and its applications
3. Apply Neural Networks and Genetic Algorithms to different problem areas
4. Use the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic.
5. Evaluate and compare solutions by various soft computing approaches for a given

problem.